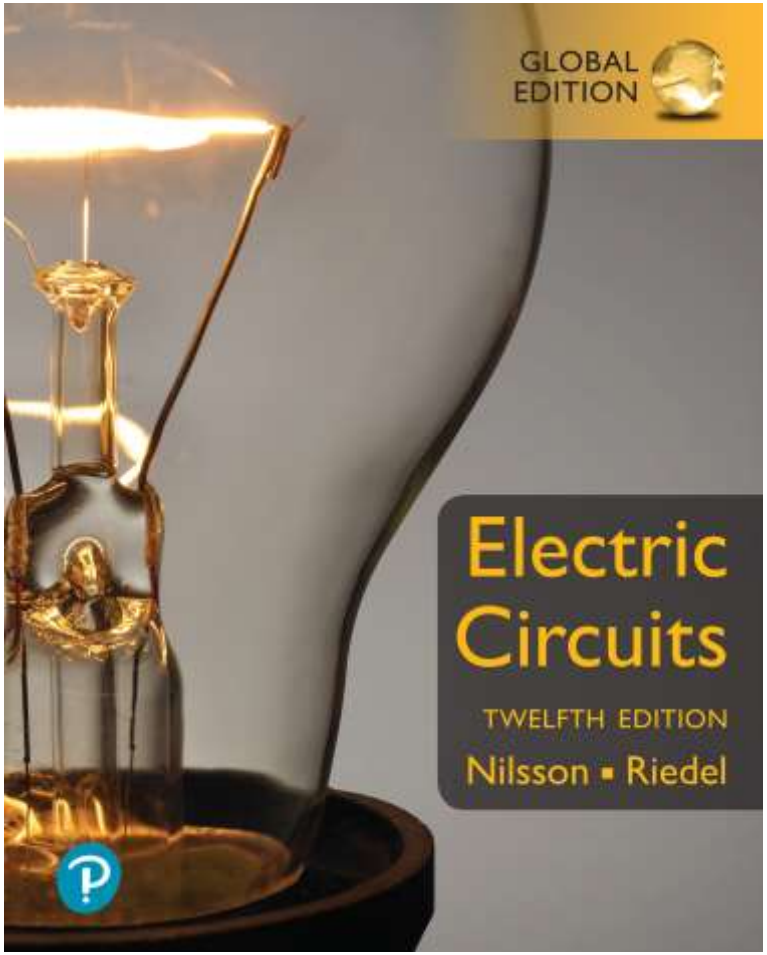
Electric Circuits

Twelfth Edition, Global Edition



Chapter 4 – Part 2

Techniques of Circuit Analysis

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

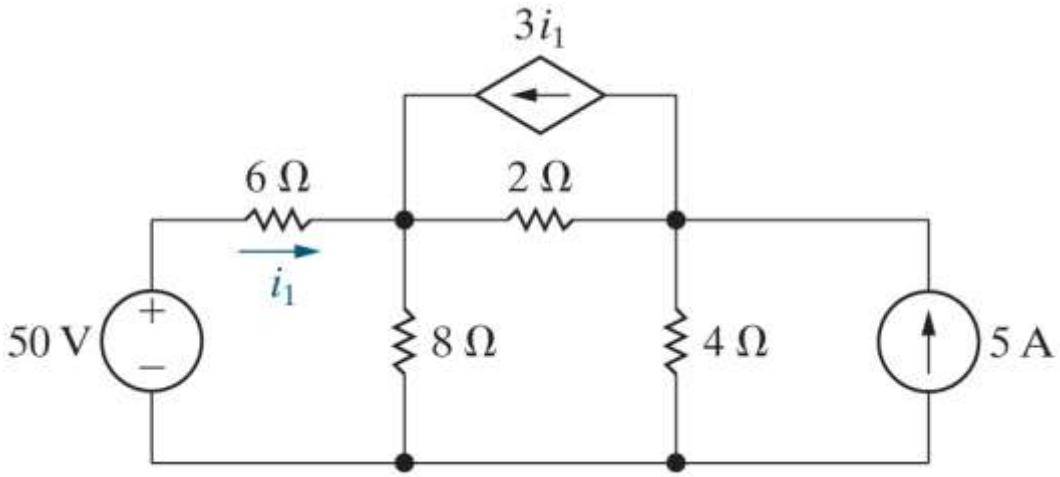
Chapter 4 – techniques of circuit analysis (1 of 26)

- Node voltage method – special cases:
  - The circuit contains one or more **dependent sources**
  - The circuit has a branch that connects **a non-reference essential node and the reference node,** and **this branch contains a voltage source only.**
  - The circuit has a branch that connects two **non-reference essential nodes**, and this **branch** contains a **voltage source only**.

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (2 of 26)

- Node voltage method with dependent sources:
  - Find the power associated with each source.

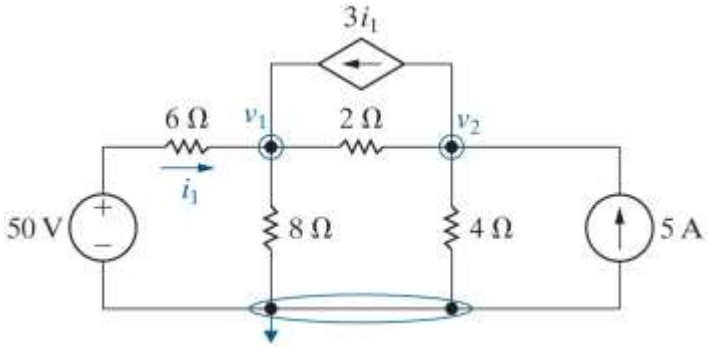


Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (3 of 26)

- Node voltage method with dependent sources:
  - Step 1 – Identify each essential node.
  - Step 2 – Pick and label a reference node; then label the node voltages at the remaining essential nodes.
  - Step 3 – Write a KCL equation for every nonreference essential node.



KCL at v_1: $\frac{v_1 - 50}{6} + \frac{v_1}{8} + \frac{v_1 - v_2}{2} - 3i_1 = 0$

KCL at v_2: $\frac{v_2 - v_1}{2} + \frac{v_2}{4} - 5 + 3i_1 = 0$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (4 of 26)

The two node voltage equations we just constructed are given below. How many unknowns do these equations contain?

$$\frac{v_1 - 50}{6} + \frac{v_1}{8} + \frac{v_1 - v_2}{2} - 3i_1 = 0$$

$$\frac{v_2 - v_1}{2} + \frac{v_2}{4} - 5 + 3i_1 = 0$$

- ✗ A. 2
- ✓ B. 3
- ✗ C. 4
- ✗ D. None of the above

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (5 of 26)

Node voltage method with dependent sources:

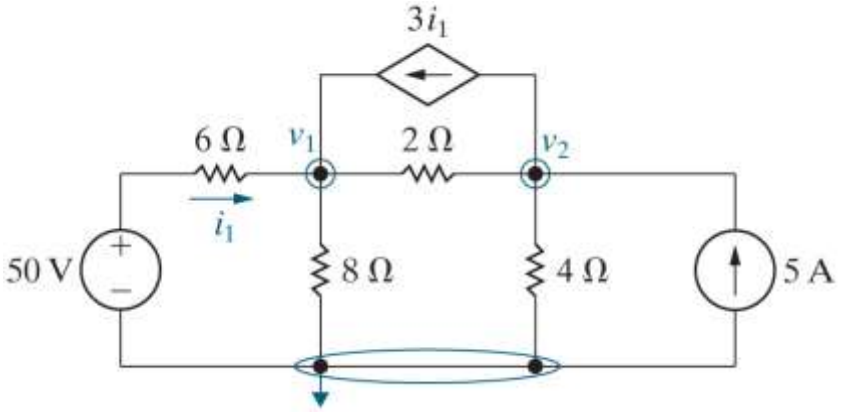
- Need to modify the analysis method:
  - Step 1 – Identify each essential node.
  - Step 2 – Pick and label a reference node; then label the node voltages at the remaining essential nodes.
  - Step 3 – Write a KCL equation for every nonreference essential node.
    - If there are dependent sources, write a constraint equation for each one.
  - Step 4 – Solve the equations to find the node-voltage values.
  - Step 5 – Solve the circuit using node voltages from Step 4.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (6 of 26)

- Node voltage method with dependent sources:
  - Step 3, continued
    - If there are dependent sources, write a constraint equation for each one.
  - Step 4 – Solve the equations to find the node-voltage values.



Pearson
© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (7 of 26)

Note that the constraint equation defines the controlling variable of the dependent source in terms of the node voltages.

KCL at v_1: $\frac{v_1 - 50}{6} + \frac{v_1}{8} + \frac{v_1 - v_2}{2} - 3i_1 = 0$ KCL at v_2: $\frac{v_2 - v_1}{2} + \frac{v_2}{4} - 5 + 3i_1 = 0$

dep. source constraint: $i_1 = \frac{50 - v_1}{6}$

Std form:

$$v_1(1/6 + 1/8 + 1/2) \quad + v_2(-1/2) \quad + i_1(-3) = 50/6$$
$$v_1(-1/2) \quad + v_2(1/2 + 1/4) \quad + i_1(3) = 5$$
$$v_1(1/6) \quad + v_2(0) \quad + i_1(1) = 50/6$$

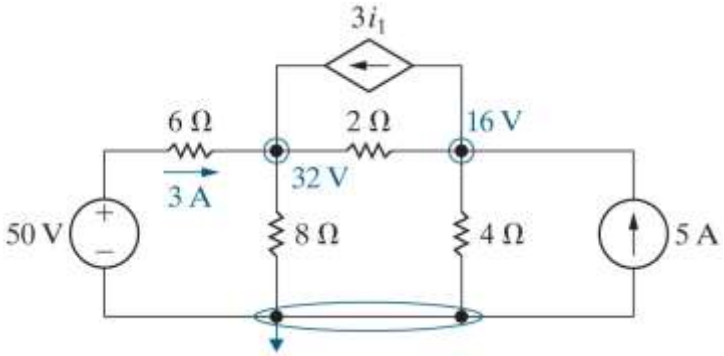
Solving: $v_1 = 32$ V; $v_1 = 16$ V; $i_1 = 3$ A

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (8 of 26)

Step 5 – Solve the circuit using node voltages from Step 4. From the power balance, we find the power for each source.



Component	Equation	p[W]
50 V	−(3)(50)	**−150**
5 A	−(5)(16)	**−80**
Dep. Source	−[3(3)](32 – 16)	**−144**
6 Ω	$(3)^2(6)$	54
2 Ω	$(32-16)^2/2$	128
8 Ω	$(32)^2/8$	128
4 Ω	$(16)^2/4$	64

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (9 of 26)

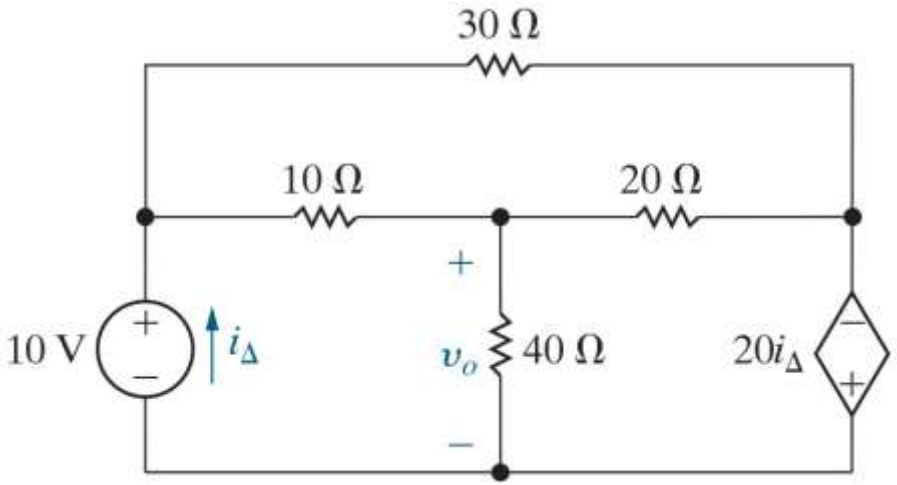
In the circuit we just analyzed, all of the sources are delivering power to the circuit.

- ✔ A. True
- ✘ B. False

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (10 of 26)

- Node voltage method with a voltage source alone in a branch:
  - Find the voltage v_o.

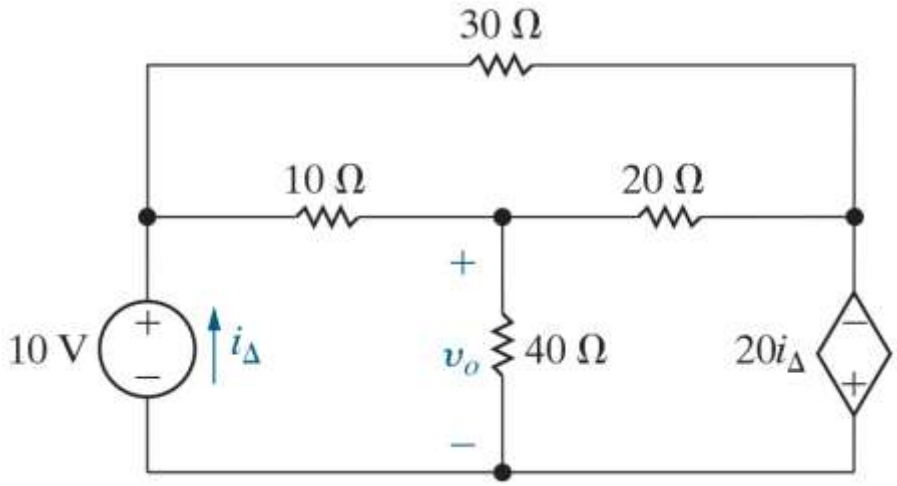


Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (11 of 26)

How many essential nodes does this circuit contain?



A. 2

B. 3

C. 4

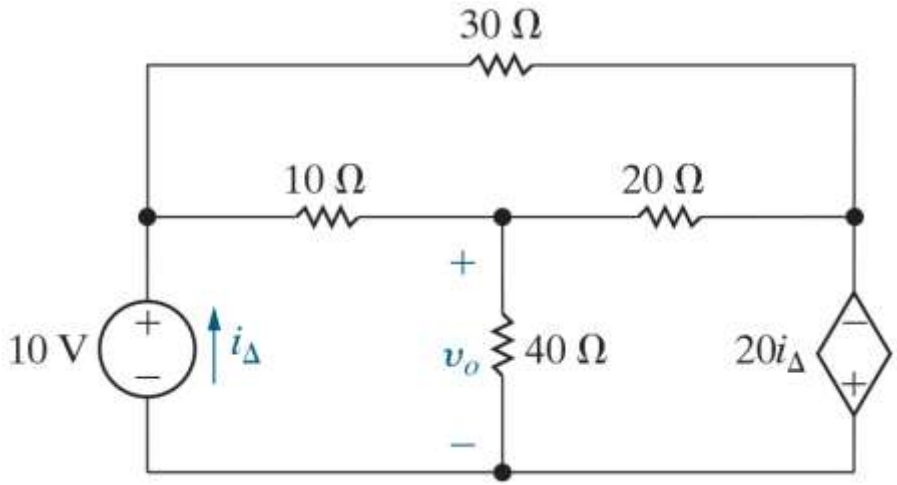
D. 5

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (11 of 26)

How many essential nodes does this circuit contain?



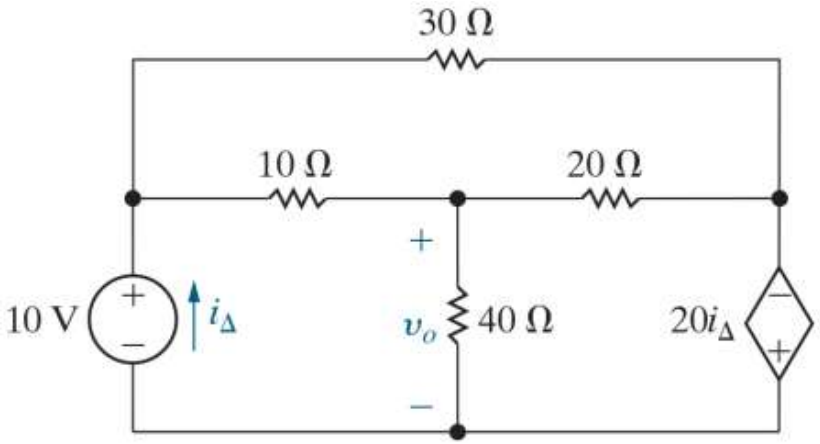
- ✗ A. 2
- ✗ B. 3
- ✓ C. 4
- ✗ D. 5

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (12 of 26)

The dependent source constraint equation must define the value of which variable?



A. i_Δ

B. v_o

C. The voltage across the dependent source

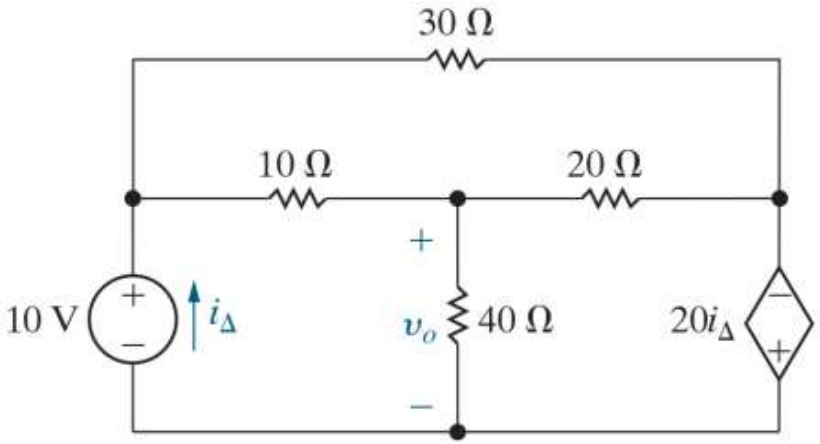
D. The reference voltage

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (12 of 26)

The dependent source constraint equation must define the value of which variable?



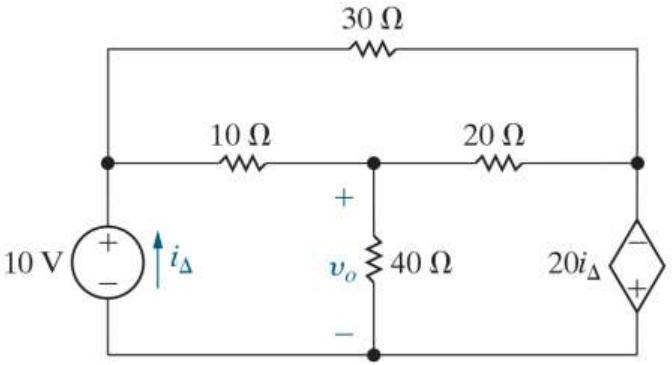
- ✔ A. i_Δ
- ✘ B. v_o
- ✘ C. The voltage across the dependent source
- ✘ D. The reference voltage

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (13 of 26)

If the essential node at the bottom of the circuit is the reference node, what is the voltage at the middle left node with respect to the reference node?



A. $20i_\Delta$

B. v_o

C. 10 V

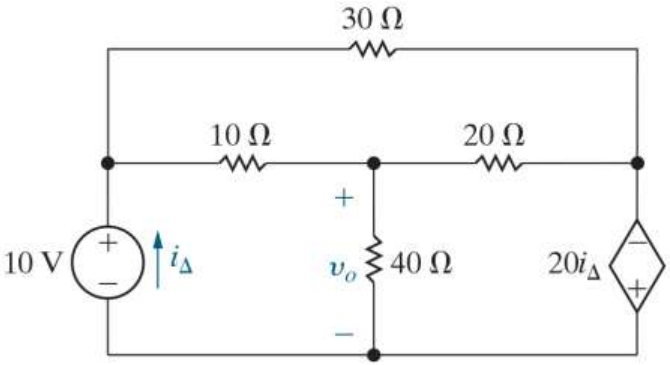
D. It is unknown

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (13 of 26)

If the essential node at the bottom of the circuit is the reference node, what is the voltage at the middle left node with respect to the reference node?



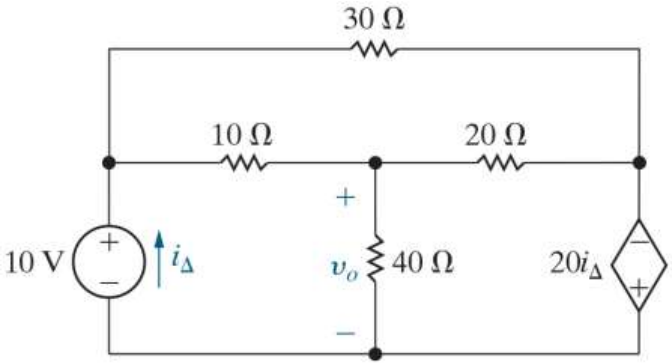
- ✗ A. $20i_\Delta$
- ✗ B. v_o
- ✓ C. 10 V
- ✗ D. It is unknown

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (14 of 26)

If the essential node at the bottom of the circuit is the reference node, what is the voltage at the middle right node with respect to the reference node?



A. $20i_\Delta$

B. v_o

C. 10 V

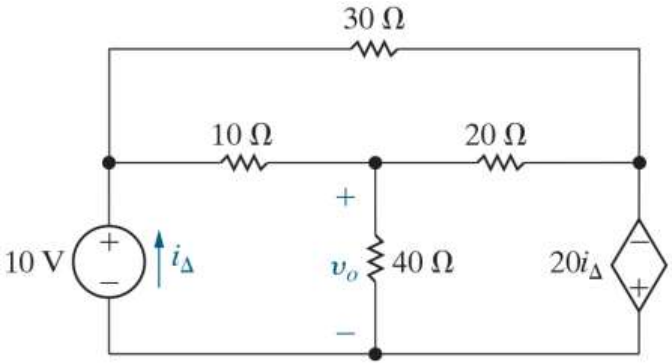
D. $-20i_\Delta$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (14 of 26)

If the essential node at the bottom of the circuit is the reference node, what is the voltage at the middle right node with respect to the reference node?



- ✗ A. $20i_\Delta$
- ✗ B. v_o
- ✗ C. 10 V
- ✓ D. $-20i_\Delta$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (15 of 26)

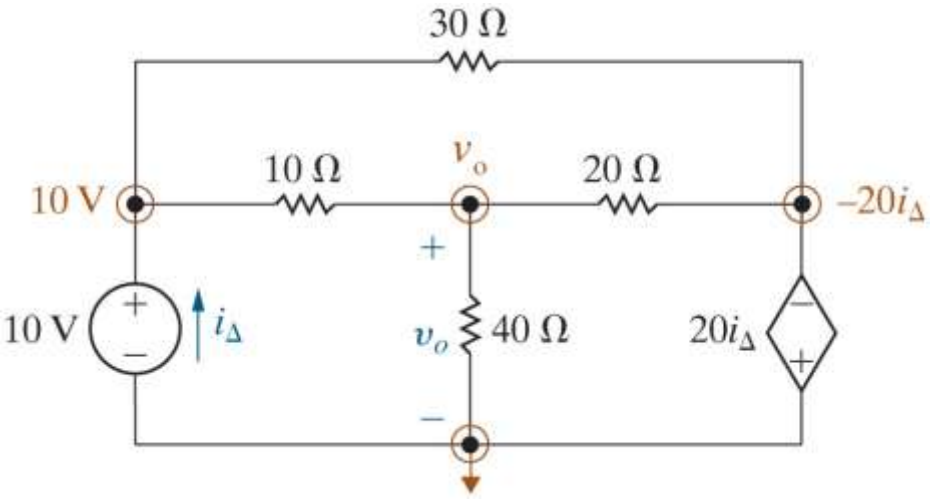
- Node voltage analysis method, modified again:
  - Step 1 – Identify each essential node.
  - Step 2 – Pick and label a reference node; then label the node voltages at the remaining essential nodes.
    - If a voltage source is the only element between an essential node and the reference node, replace the node voltage label with the value of the voltage source.
  - Step 3 – Write a KCL equation for every nonreference essential node where the voltage is unknown.
    - If there are dependent sources, write a constraint equation for each one.
  - Step 4 – Solve the equations to find the node-voltage values.
  - Step 5 – Solve the circuit using node voltages from Step 4.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (16 of 26)

- Node voltage method with a voltage source alone in a branch:
  - After Steps 1 and 2:



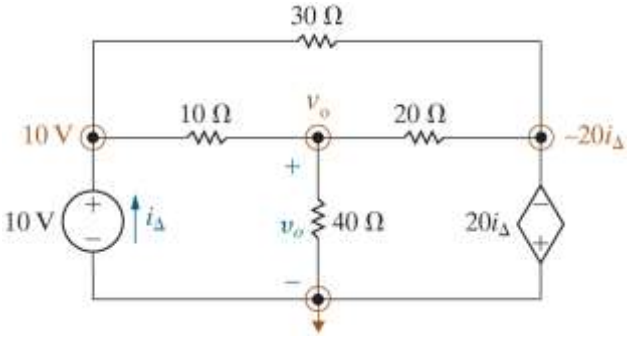
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (17 of 26)

- Node voltage method with a voltage source alone in a branch:
  - Step 3 (write the KCL and constraint equations) and Step 4 (solve the equations):

At v_o : $\dfrac{v_o - 10}{10} + \dfrac{v_o}{40} + \dfrac{v_o + 20i_\Delta}{20} = 0$



constraint: $i_\Delta = \dfrac{10 - v_o}{10} + \dfrac{10 + 20i_\Delta}{30}$

Std. form:

$$v_o\left(\frac{1}{10} + \frac{1}{40} + \frac{1}{20}\right) + i_\Delta\left(\frac{20}{20}\right) = \frac{10}{10}$$

$$v_o\left(-\frac{1}{10}\right) + i_\Delta\left(1 - \frac{20}{30}\right) = \frac{10}{10} + \frac{10}{30}$$

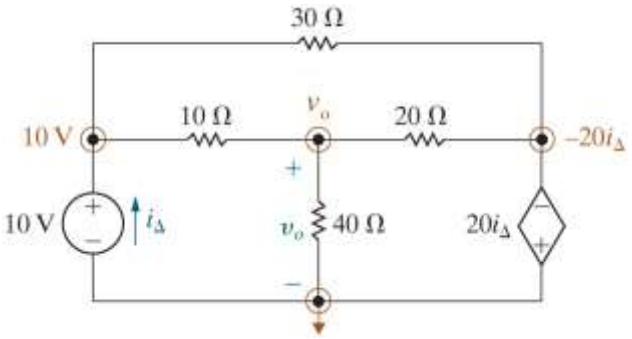
Solution: $v_o = 24$ V; $i_\Delta = -3.2$ A

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (18 of 26)

Power balance:



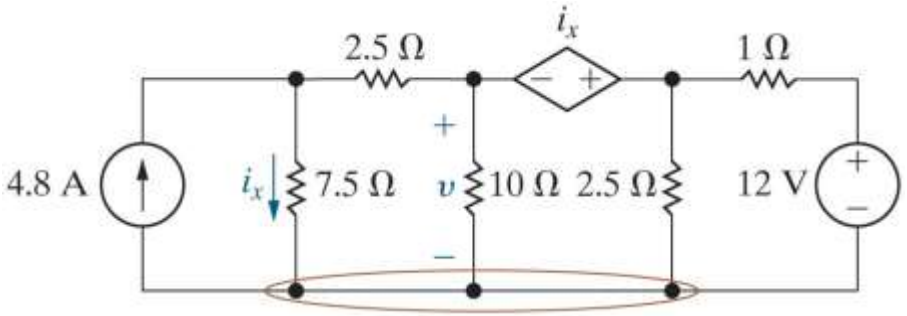
Component	Equation	p [W]
10 V	−(10)(−3.2)	32
Dep. Source	−[−3.2 – 24/40][20(−3.2)]	−243.2
30 Ω	$[-20(-3.2)-10]^2 / 30$	97.2
10 Ω	$(24-10)^2 / 10$	19.6
20 Ω	$[24 + 20(-3.2)]^2 / 20$	80
40 Ω	$24^2 / 40$	14.4

Pearson

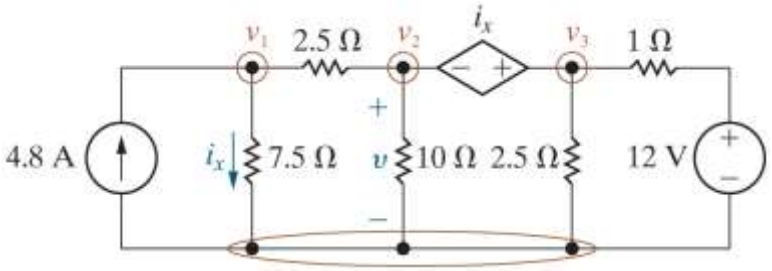
© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (19 of 26)

- Node voltage method with a voltage source between two non-reference essential nodes:
  - Find the voltage v.



After Steps 1 and 2,

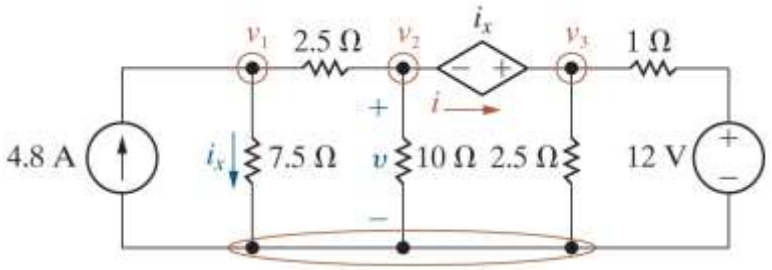


Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (20 of 26)

- Node voltage method with a voltage source between two non-reference essential nodes:
  - Step 3 – Write the KCL equations. To do this, introduce a 4th unknown, *i*.



$$\text{At } v_1: -4.8 + \frac{v_1}{7.5} + \frac{v_1 - v_2}{2.5} = 0$$

$$\text{At } v_2: \frac{v_2 - v_1}{2.5} + \frac{v_2}{10} + i = 0$$

$$\text{At } v_3: -i + \frac{v_3}{2.5} + \frac{v_3 - 12}{1} = 0$$

Sum the last two equations to eliminate *i*:

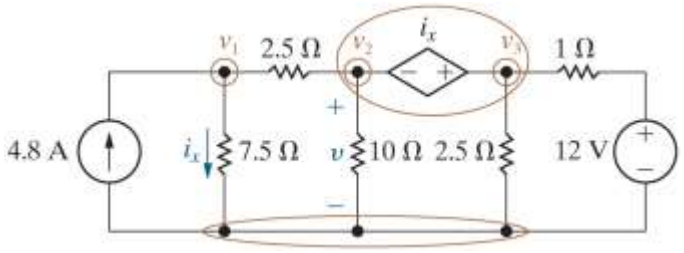
$$\frac{v_2 - v_1}{2.5} + \frac{v_2}{10} + i - i + \frac{v_3}{2.5} + \frac{v_3 - 12}{1} = 0!$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (21 of 26)

- Node voltage method with a **supernode**:
  - **Instead of introducing another unknown**, combine the two essential nodes and the voltage source in the branch between them into a single supernode. Then in Step 3, write a KCL equation for the supernode and a constraint equation for the supernode.



At v_1: $-4.8 + \frac{v_1}{7.5} + \frac{v_1 - v_2}{2.5} = 0$

At supernode: $\frac{v_2 - v_1}{2.5} + \frac{v_2}{10} + \frac{v_3}{2.5} + \frac{v_3 - 12}{1} = 0$

SN constraint: $v_3 - v_2 = 1i_x$

Dep source constraint: $i_x = \frac{v_1}{7.5}$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (22 of 26)

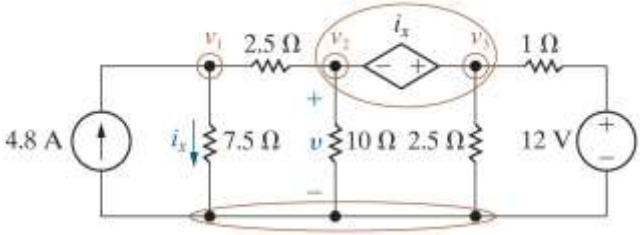
- Node voltage analysis method, final version
  - Step 1 – Identify each essential node.
  - Step 2 – Pick and label a reference node; then label the node voltages at the remaining essential nodes.
    - If a voltage source is the only element between an essential node and the reference node, replace the node voltage label with the value of the voltage source.
    - **If a voltage source is the only element between two nonreference essential nodes**, combine the nodes and the source into a single supernode.
  - Step 3 – Write a KCL equation for each supernode and for every remaining nonreference essential node where the voltage is unknown.
    - If there are dependent sources, write a constraint equation for each one.
    - **If there are supernodes, write a constraint equation for each one.**
  - Step 4 – Solve the equations to find the node-voltage values.
  - Step 5 – Solve the circuit using node voltages from Step 4.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (23 of 26)

- Node voltage method with a super node:
  - Find the voltage v.



Step 4 – Solve the equations (using standard form):

$$v_1\left(\frac{1}{7.5}+\frac{1}{2.5}\right)+v_2\left(-\frac{1}{2.5}\right)+v_3(0)+i_x(0)=4.8$$

$$v_1\left(-\frac{1}{2.5}\right)+v_2\left(\frac{1}{2.5}+\frac{1}{10}\right)+v_3\left(\frac{1}{2.5}+\frac{1}{1}\right)+i_x(0)=12$$

$$v_1(0)+v_2(-1)+v_3(1)+i_x(-1)=0$$

$$v_1\left(-\frac{1}{7.5}\right)+v_2(0)+v_3(0)+i_x(1)=0$$

$v_1 = 15$ V; $\boxed{v = v_2 = 8 \text{ V};}$ $v_3 = 10$ V; $i_x = 2$ A

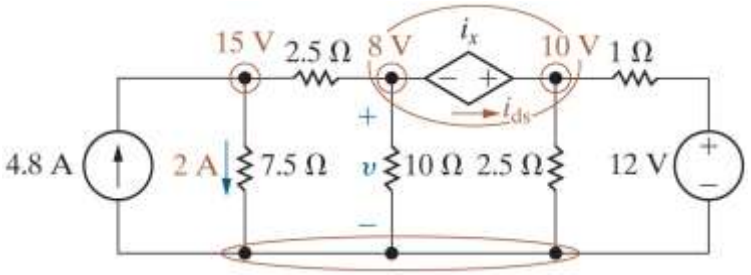
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (24 of 26)

Power balance:

$$i_{ds} = \frac{10}{2.5} + \frac{10-12}{1} = 2 \text{ A}$$



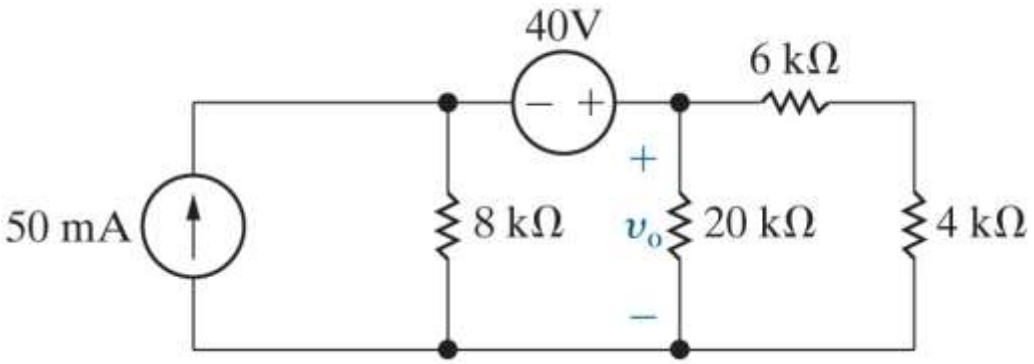
Component	Equation	p [W]
4.8 A	−(4.8)(15)	−72
Dep. Source	−(2)(2)	−4
12 V	(12)(10 – 12)/1	−24
7.5 Ω	$(2)^2(7.5)$	30
2.5 Ω (top)	$(15-8)^2/2.5$	19.6
10 Ω	$(8)^2/10$	6.4
2.5 Ω	$(10)^2/2.5$	40
1 Ω	$(10-12)^2/1$	4

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (25 of 26)

How many super nodes does this circuit contain?



A. 0

B. 1

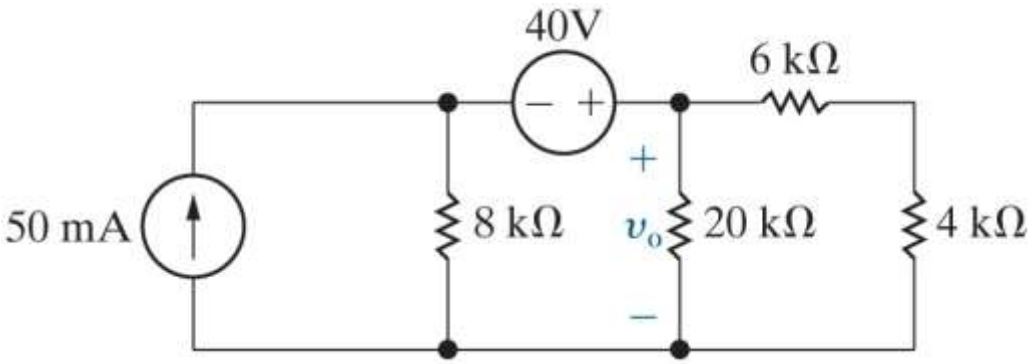
C. 2

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (25 of 26)

How many super nodes does this circuit contain?



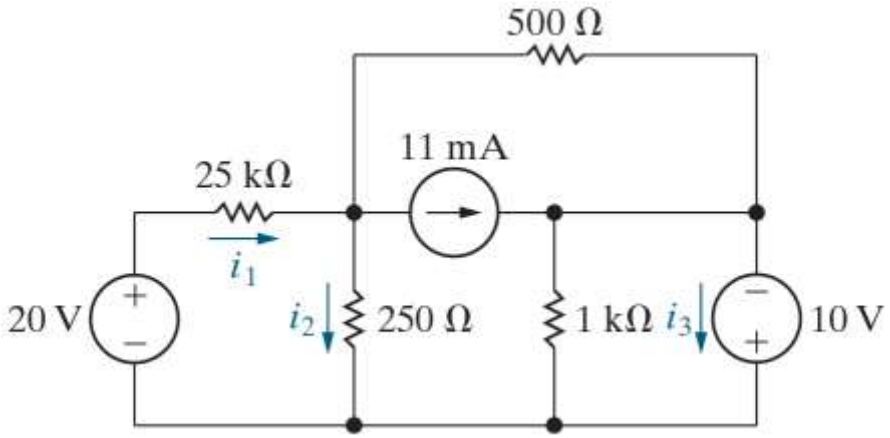
- ✗ A. 0
- ✓ B. 1
- ✗ C. 2

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (26 of 26)

How many super nodes does this circuit contain?



A. 0

B. 1

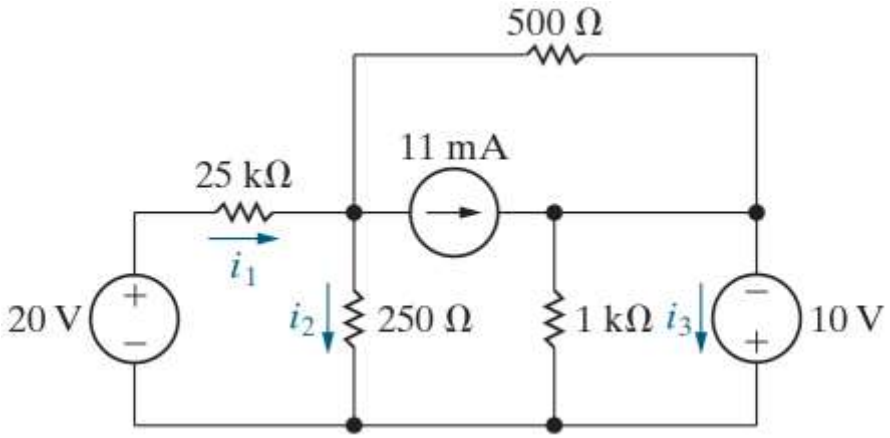
C. 2

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 4 – techniques of circuit analysis (26 of 26)

How many super nodes does this circuit contain?



- ✔ A. 0
- ✘ B. 1
- ✘ C. 2

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Copyright



This work is protected by United Kingdom copyright laws and is provided solely for the use of instructors in teaching their courses and assessing student learning. Dissemination or sale of any part of this work (including on the World Wide Web) will destroy the integrity of the work and is not permitted. The work and materials from it should never be made available to students except by instructors using the accompanying text in their classes. All recipients of this work are expected to abide by these restrictions and to honor the intended pedagogical purposes and the needs of other instructors who rely on these materials.

© 2024 Pearson Education Ltd. All Rights Reserved.